# Ravindra **Yadav, Ph.D.**

Senior Machine Learning Engineer | Generative AI • LLMs • Agentic AI • Multimodal AI

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## Professional Summary

Senior Machine Learning Engineer and Ph.D. with 8+ years of experience designing, developing, and productionizing AI/ML systems at the intersection of Generative AI, LLMs, AI Agents, multimodal learning, computer vision, and edge AI. Currently leading an 8-member engineering team at Samsung R&D, owning the full ML lifecycle from model research and experimentation to system architecture and commercialization. Experienced in building scalable, production-grade ML systems serving millions of user sessions worldwide.

## Work Experience

### Samsung R&D Institute

*Delhi, India*

Chief Engineer — *March 2022 - Present*

- Lead an 8-member AI/ML engineering team delivering production AI capabilities across Samsung Smart TVs and gaming monitors, taking ML systems from research and prototyping through deployment and commercialization, powering experiences across millions of user sessions daily worldwide.
- Architected and delivered an agentic gaming personalization platform that dynamically builds user preference profiles, orchestrates Q&A-driven interactions, and generates context-aware game recommendations; processed 10K+ gameplay signals/day, achieving 95%+ recommendation relevance and driving a 35–40% increase in recommendation engagement.
- Led development of a production Retrieval-Augmented Generation (RAG) platform for enterprise knowledge access, indexing 3K+ documents/knowledge artifacts with 0.73 MRR@5; achieved 90%+ Precision@5/Recall@5, and reduced knowledge-search time by 60–70% compared with manual information retrieval.
- Developed an LLM-powered defect analysis platform that automatically interprets error logs, performs root-cause analysis, and recommends solutions; processed 1K+ defect reports/logs, achieving 80% root-cause identification accuracy and reducing manual debugging effort by 70%.
- Developed an on-device sound event detection system for indoor speech and environmental sounds, achieving 97%+ F1 score at <100 ms inference latency while operating within 5 MB memory constraints; enabled real-time notifications and automated daily-event summarization without requiring continuous cloud inference.
- Designed and productionized Autonomous Interaction, a vision-language model (VLM) based interface enabling end-to-end TV interaction—including UI navigation, application launching, and content playback—with 90%+ task-completion rate, <1.5 s median response latency, and support for 10+ OTT apps.

### TCS Innovation Lab

*Kolkatta, India*

Research Intern — *May 2021 - Nov 2021*

- Title: Cognitive Robotics - Personalised avatar and motion capture
- Developed a one-shot, facial geometry-aware emotional talking face generation system capable of generalizing across diverse identities.
- Addressed limitations of existing AI-based talking face generation methods by leveraging a Graph Convolutional Network (GCN) on facial landmark graphs to drive speech and emotion while preserving facial structure.

### NVIDIA

*Bangalore, India*

Research Intern — *Feb. 2019 - Oct 2020*

- Title: Generative models for multimodal representation learning
- Developed MVAE-LP, a multimodal variational autoencoder framework to enhance representation learning across diverse modalities.
- Solved limitations of existing multimodal generative models—such as over-regularization, poor scalability, and multi-step divergence—resulting in improved predictive performance on bimodal, trimodal, and quadrimodal datasets.

### Vinculum Solutions Pvt Limited

*Delhi, India*

Siebel Administrator — *Mar. 2016 - Aug 2016*

- Served as a Siebel Server Administrator, responsible for managing CRM server operations, performing system upgrades, and ensuring optimal server performance and uptime.

### Indraprastha Institute of Information Technology (IIIT)

*Delhi, India*

Research Assistant — *Jan 2016 – March 2016*

- Title: Development and Implementation of an Automated System for Counterfeit Currency Detection
- Developed an automated computer vision system for counterfeit currency detection using region-based feature analysis.
- Addressed the challenge of accurately identifying counterfeit notes by processing distinct note features independently and aggregating regional predictions to improve detection reliability.

**MPHASIS** *Pune, India*

Senior Specialized Infrastructure Engineer *Jan. 2009 - July 2012*

- As a Middleware Engineer, managed the installation, configuration, and maintenance of server environments across diverse platforms, ensuring seamless integration and system stability. Collaborated closely with development teams to diagnose and resolve daily operational issues, driving efficient and timely solutions.

## Technical Skills

**Core areas:** AI agents, Large language models (LLMs), Retrieval augmented generation (RAG), Computer vision, Generative AI (GenAI), Avatar generation, Speech generation, Anomaly detection

**ML Frameworks:** Pytorch | Tensorflow | ClaudeCode | Huggingface | LangGraph | LangChain | OpenCV

**Languages:** Python, C, C++, JAVA, HTML, CSS, Java script

## Education

### Doctor of Philosophy (PhD)

*IIT Kanpur*

Electrical Engineering *Jan. 2017 - Mar 2022*

- Title: Understanding Multimodal Representation Learning via Latent Variable Models
- Advisors: Prof. Vinay P Namboodiri and Prof. Rajesh M Hegde
- CGPA: 8.0/10

### Master of Technology

*IIT Kanpur*

Electrical Engineering *Aug. 2012 - Sep. 2014*

- Title: A Unified Approach for Archival Film Restoration using Video Decomposition
- Advisor: Prof. Sumana Gupta
- CGPA: 7.25/10

### Bachelor of Engineering

*Govt. Engg. College Ajmer*

Electronics and Communication Engineering *Aug. 2004 - Aug. 2008*

- Percentage: 70%

## Publications

### Journals:

- Ravindra Yadav, Saumik Bhattacharya, K. S. Venkatsh, and Sumana Gupta. "A spatiotemporal restoration of partial color artifacts in archival films." Signal, Image and Video Processing (2016): 1-8. url: https://link.springer.com/article/10.1007/s11760-016-0945-y

### Conferences:

- Ravindra Yadav, Ashish Sardana, Vinay P Namboodiri, Rajesh M Hegde. "Learning to Predict Speech in Silent Videos via AudioVisual Analogy." In International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2022.
- Ravindra Yadav, Munender Varshney, Vinay P Namboodiri, Rajesh M Hegde. "Learning Speaker Specific Lip to Speech Generation." In International Conference on Pattern Recognition (ICPR) 2022.
- Sandika Biswas, Sanjana Sinha, Ravindra Yadav, Brojeshwar Bhowmick. "Emotion-Controllable Generalized Talking Face Generation." In International Joint Conference on Artificial Intelligence (IJCAI) 2022.
- Ravindra Yadav, Ashish Sardana, Vinay P Namboodiri, Rajesh M Hegde. "Speech Prediction in Silent Videos using Variational Autoencoders." In International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2021
- Ravindra Yadav, Ashish Sardana, Vinay P Namboodiri, Rajesh M Hegde. Stochastic Talking Face Generation Using Latent Distribution Matching. In Proceedings of Interspeech 2020
- Ravindra Yadav, Ashish Sardana, Vinay P Namboodiri, Rajesh M Hegde. "Bridged Variational Autoencoders for Joint Modeling of Images and Attributes." Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2020, pp. 1479-1487
- Ravindra Yadav, Bhattacharya, Saumik, V. Narendra, K. S. Venkatsh, and Sumana Gupta. "Total variation based fast video decomposition for artifact restoration." In International Conference on Digital Signal Processing (DSP), pp. 1152-1155. IEEE, 2015.

## Academic Services/Achievements

- Served as a reviewer for premier machine learning and computer vision conferences, including CVPR, ICCV, and ECCV, contributing to the evaluation of cutting-edge research in the field.
- Achieved a top 0.3% ranking among 176,944 candidates in the Graduate Aptitude Test in Engineering (GATE) 2012.
- Recruited by Bharat Electronics Limited, one of India's prestigious Navratna public sector enterprises, in 2010.